

# Brice Peres

PhD candidate – Computer Graphics & Physical Simulation

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Portfolio: [briceprs.github.io](https://briceprs.github.io) ↗



## Experience

### Inria ELAN team – PhD candidate

*Numerical study of naturally-curved fibre assemblies*

Apr. 2026 – present

Grenoble, France

- Numerical modelling and simulation of bundles of naturally-curved fibres (continuation of the work below).

### Inria ELAN team – Pre-PhD researcher

*Tomographic hair-data filtering*

Sept. 2025 – Mar. 2026

Grenoble, France

- Designed filtering methods for tomographic acquisitions of human hair, in preparation of the PhD project.

### Dassault Systèmes – End-of-studies intern

*3D engine performance optimisation*

Sept. 2024 – Mar. 2025

Paris, France

- Worked with the rendering team to improve the performance of one of Dassault Systèmes' main 3D engines.

### Corys – R&D intern

*Unreal Engine 5, runtime train simulator*

May 2023 – Sept. 2023

Grenoble, France

- Implemented real-time windscreen effects (rain, snow, dirt, cleaning) on Unreal Engine 5 (HLSL).
- Worked in the R&D team led by Alain Kocelniack.

### Inria – Research intern

*GPU terrain filtering*

June – July 2022

Grenoble, France

- Designed a GPU height-map filtering algorithm for terrain generalisation (advisors: Joëlle Thollot, Romain Vergne).

## Selected projects

### Real-time soft-body simulation

*C++, CUDA*

Spring-mass soft body accelerated on the GPU with CUDA.

Dec. 2024

### Progressive raytracer

*C++, OpenGL*

GPU raytracer with BVH traversal, shadows, reflections, refractions and environment maps. Cook–Torrance shading with a GGX microfacet distribution; frame accumulation for progressive refinement.

Dec. 2024

### Particle physics simulation

*C++*

2D/3D simulation under gravity and a Lennard-Jones potential with custom spatial data structures. Ran 8 000 particles over 400 000 steps in 20 minutes on a laptop.

May 2023

### Black Seas – multiplayer naval game

*Rust, OpenGL – team of 5*

From-scratch engine; rendering, terrain, lighting, post-processing (ray-marched clouds, bloom, tonemapping).

Feb. – Apr. 2023

## Education

### ENSIMAG – Engineering degree (M.Eng.), Grenoble INP

*Computer science and applied mathematics*

2021 – 2024

Grenoble, France

### Institute of Science Tokyo (formerly Tokyo Tech)

*Academic exchange semester*

Mar. – Aug. 2024

Tokyo, Japan

### Lycée Notre-Dame de Sion – Classes préparatoires MPSI/MP

*Two-year intensive course for French engineering school exams*

2018 – 2021

Marseille, France

## Skills, languages & interests

**Languages.** C++, Rust, C, Python, GLSL, HLSL

**Graphics & GPU.** OpenGL, CUDA, WebGL, Unreal Engine 5, Shadertoy

**Tools.** Git,  $\LaTeX$ , Linux • **Familiar.** Java, Shell, Assembly

**Spoken.** French (native) • English (C1, TOEFL 100 pts) • Japanese (A1)

**Interests.** Climbing & bouldering since 2011 (a few competitions); hiking; computer graphics, physically based rendering, real-time simulation, Shadertoy.